Landslide Exposure Database: Codebase & Workflow

This document outlines the architecture, scripts, and step-by-step workflow used to generate the **US Landslide Exposure Database**. The pipeline is divided into three main stages: Data Acquisition (`0_input`), Data Processing & Integration (`1_process`), and Analytics & Visualization (`2_output`).

Recommended Python Environment

To ensure full compatibility with the spatial and data processing libraries used in these scripts, it is highly recommended to use the **HyRiver** Python environment.

You can download the environment configuration file here: 🔗 hyriver/environment.yml

To set up the environment using Conda or Mamba:

```
# Download the environment.yml file, then run:
conda env create -f environment.yml
conda activate hyriver
```

📂 Repository Structure

0_input : Data Acquisition

This directory contains the scripts responsible for fetching the foundational building, hazard, and demographic datasets.

- **Script 1: Download_building_data.ipynb**
  - Fetches raw building footprint geometries from Overture Maps.
  - Retrieves structural and occupancy attributes from the National Structure Inventory (NSI).
- **Script 2: Download_census_data.ipynb**
  - Downloads foundational geographic boundaries including: Physiographic Provinces, Physiographic Divisions, US States, Census Tracts, and Census Blocks.
  - Fetches socioeconomic and demographic datasets:
    - **CRE:** Community Resilience Estimates.
    - **ACS:** American Community Survey (Demographics, Poverty, Income).
    - **FFIEC:** Federal Financial Institutions Examination Council (FFIEC) income level data.
- **Script 3: Download_USGS_landslideSusc_n10.ipynb**
  - Retrieves USGS N10 Landslide Susceptibility Model.

1_process : Data Processing and Integration

This directory contains the core analytical engine of the project, merging spatial data, applying machine learning filters, and aggregating metrics. The scripts follow a sequential order, in which the worflow goes from a to e.

- **Script 4: a_LandslideExposureDatabase_part1.ipynb**
  - **Part 1: Building Inventory Integration**
    - *Spatial Comparison:* Analysis to perform spatial join and nearest-neighbor.
    - *Geometry & Proximity Filters:* Cleaning and aligning raw footprint data.
    - *Machine Learning:* Implementation of the Decision Tree classifier to distinguish primary structures (e.g., houses) from secondary outbuildings (e.g., detached garages).
    - *Landslide Susceptibility Attribution:* Intersecting building footprints with hazard rasters.
- **Script 5: b_LandslideExposureDatabase_part2.ipynb**
  - **Part 2: Geographic Classification**
    - Spatially joining buildings to their respective Physiographic Divisions, Census Divisions, States, Census Tracts, and Census Blocks.
- **Script 6: c_LandslideExposureDatabase_part2.ipynb**
  - **Part 3: Socioeconomic Attributes**
    - Assigning population estimates to buildings (daytime/nighttime).
    - Incorporating error metrics, Community Resilience Estimates (CRE), and poverty/income data at the structural level.
- **Script 7: d_CensusTractsMaster.ipynb**
  - **Building Aggregation:** Rolling up structure-level landslide exposure metrics to the Census Tract level.
  - **Socioeconomic Aggregation:** Aggregating CRE, poverty, and income data to Census Tracts.
- **Script 8: e_LISA_cluster_LED-CRE.ipynb**
  - **LISA Clusters:** Calculating Local Indicators of Spatial Association (LISA) to identify statistically significant clusters of exposed and vulnerable populations.

📊 2_output : Analytics and Visualization

This directory contains the scripts used to generate the final analytical figures, maps, and plots for publication.

- **Script 9: Results_figures.ipynb**
  - National Landslide-Susceptibility Exposure Assessment. Generates all visual assets based on the Master Database, including:
    - **National Exposure Maps:** High and Moderate-to-High susceptibility areas.
    - **Geographic Summaries:** State-level and Tract-level exposure plots.
    - **Regional Breakdowns:** Analytics sliced by Physiographic Divisions and Census Divisions.
    - **Socioeconomic Characteristics:** Plots correlating landslide exposure with Income, Poverty, and LISA cluster results.